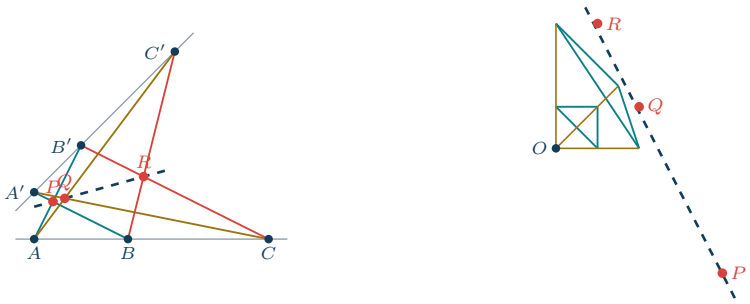
PROJECTIVE GEOMETRY LADDER 6 OF 9

Pappus and Desargues by cross products

Two theorems about lines and meeting points become exact identities once points at infinity are allowed.

1 / THE RULE AND THE PICTURE

Six points on two lines give three meeting points on one line



Pappus: take A, B, C on one line and A', B', C' on another, then the points $AB' \cap A'B$, $AC' \cap A'C$ and $BC' \cap B'C$ are collinear.
Desargues: triangles ABC and $A'B'C'$ whose corresponding vertices are joined by concurrent lines have corresponding sides meeting on a line.

2 / PAPPUS IN COORDINATES

Every meeting point is a cross product and collinearity is a determinant

case	first line	second line	the three meeting points
finite case	$A = (0,0)$ $B = (2,0)$ $C = (5,0)$	$A' = (0,1)$ $B' = (1,2)$ $C' = (3,4)$	$P = (\frac{2}{5}, \frac{4}{5})$ $Q = (\frac{15}{23}, \frac{20}{23})$ $R = (\frac{2}{3}, \frac{4}{3})$ $\det(P,Q,R) = 0$
one point at infinity	$A = (0,0)$ $B = (1,0)$ $C = (3,0)$	$A' = (0,-1)$ $B' = (1,1)$ $C' = (2,3)$	$P = (1:1:0)$ $Q = (-\frac{6}{7}, -\frac{9}{7})$ $R = (\frac{9}{7}, \frac{6}{7})$ collinear along slope 1

3 / WHEN A MEETING POINT IS AT INFINITY

The theorem still holds

In the second case the lines AB' and $A'B$ both have slope 1, so in the plane they have no common point and the statement would have an exception.
Projectively they meet at $P = (1:1:0)$, and the line through Q and R has slope 1, so it passes through P and the three points are collinear.
The computer check uses exact fractions and finds the determinant exactly zero.

4 / DESARGUES IN COORDINATES

Perspective from a point means perspective from a line

Take $A, B, C = (1,0), (0,1), (1,1)$ and $A', B', C' = (2,0), (0,3), (\frac{3}{2}, \frac{3}{2})$, which lie on lines through the origin O at ratios $2, 3, \frac{3}{2}$.
The corresponding sides meet at $(4,-3)$, $(2,1)$ and $(1,3)$, and the determinant of these three points is exactly zero, so they lie on the line $y = -2x + 5$.

5 / WHY THESE ARE PROJECTIVE THEOREMS

Their statements use only lines and points

Neither theorem mentions a length or an angle, only joining and meeting, so they hold in every projective plane that comes from a field and are unchanged by any projective map.
Pascal's theorem is the sibling for six points on a conic, whose three opposite-side meeting points are collinear, and Pappus is the case where the conic splits into two lines.

6 / TRY THESE

Compute a join and a meet

(a) Join $(1:0:1)$ and $(0:1:1)$: the cross product is $[-1:-1:1]$, the line $x + y = 1$.
(b) Meet $x = 0$ and $y = 0$: $[1:0:0] \times [0:1:0] = (0,0,1)$, the origin.
(c) Three points are collinear when $\det = 0$: for $(0,0), (1,1), (2,2)$ the triples are $(0,0,1), (1,1,1), (2,2,1)$ and the determinant is 0.

ASIDE / NO MEASUREMENT NEEDED

Because only joins and meets appear, the proofs can be done with cross products and determinants and need no distances, angles or coordinates adapted to the picture.

LINKS TO OTHER SHEETS

The homogeneous coordinates sheet gives the cross product rules, and the Bézout sheet explains why a degree count forces such coincidences.

Convention. P, Q, R are the three meeting points; collinear means $\det = 0$; a point $(a:b:0)$ lies on the line at infinity; computations use exact fractions.